Measuring the Solar Mixing Angle θ_{12}

A Chi-Squared Analysis of SNO Phase I Data
in the Adiabatic MSW Regime

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Abstract

We reproduce the measurement of the solar neutrino mixing angle θ_{12} using the charged-current (CC) and neutral-current (NC) flux measurements from the Sudbury Neutrino Observatory (SNO) Phase I dataset (2002). Working in the adiabatic Mikheyev–Smirnov–Wolfenstein (MSW) regime, appropriate for high-energy ^{8}B solar neutrinos, the electron-neutrino survival probability reduces to $P_{ee} = \sin^2\theta_{12}$. We perform a chi-squared minimisation of the CC/NC flux ratio and extract

$$\theta_{12} = 36.0^\circ \pm 2.3^\circ,$$

consistent with the SNO-only result published in 2002. We demonstrate that the vacuum-averaged formula $P_{ee} = 1 - \frac{1}{2}\sin^2 2\theta_{12} \approx 0.548$ is incompatible with the observed suppression of $\sim$34%, confirming that matter effects are essential to the correct interpretation of solar neutrino data. The slight offset from the current PDG combined value of $\theta_{12} = 33.4^\circ$ is explained qualitatively through the Earth's day-night regeneration effect and the subsequent KamLAND constraint on Δm^2_{21}.

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1 Introduction

The solar neutrino problem — the deficit of electron neutrinos arriving at Earth relative to Standard Solar Model (SSM) predictions — stood as one of the most compelling anomalies in particle physics for over three decades [1, 2]. Early radiochemical experiments such as Homestake, GALLEX, and SAGE observed roughly one-third to one-half the predicted flux, but could not determine whether the missing neutrinos had oscillated into other flavours or whether the SSM itself was simply wrong.

The Sudbury Neutrino Observatory (SNO), located 2 km underground in the INCO Creighton mine in Sudbury, Ontario (now SNOLAB), resolved this ambiguity decisively [5]. By using heavy water (D_2O) as both target and Cherenkov medium, SNO could simultaneously measure:

- the **charged-current** (CC) reaction $\nu_e + d \to p + p + e^-$, sensitive only to electron neutrinos;
- the **neutral-current** (NC) reaction $\nu_x + d \to p + n + \nu_x$, equally sensitive to all active flavours.

The ratio $\phi_{\rm CC}/\phi_{\rm NC}$ directly measures the electron-neutrino survival probability P_{ee}, independent of solar model uncertainties. When SNO reported $\phi_{\rm NC} \approx \phi_{\rm SSM}$ but $\phi_{\rm CC} \approx 0.35\,\phi_{\rm SSM}$, it provided the first direct, model-independent proof that solar neutrinos change flavour in transit [5].

In this paper we reproduce the SNO Phase I extraction of θ_{12}, emphasising the critical role of the MSW matter effect. Section 2 derives the relevant survival probability from first principles. Section 3 describes the experimental inputs and chi-squared method. Section 4 presents our fit result and walks through each panel of the analysis figure. Section 5 discusses the limitations of the SNO-only measurement and the refinements brought by KamLAND and the Earth's regeneration effect.

2 Theoretical Framework

2.1 Two-Flavour Neutrino Oscillations in Vacuum

In vacuum, the flavour eigenstates $|\nu_e\rangle$ and $|\nu_\mu\rangle$ are related to the mass eigenstates $|\nu_1\rangle$ and $|\nu_2\rangle$ by the PMNS mixing matrix, which in the two-flavour approximation reduces to a single rotation by angle θ_{12}:

$$\begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_{12} & \sin\theta_{12} \\ -\sin\theta_{12} & \cos\theta_{12} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \end{pmatrix}. \tag{1}$$

For a neutrino propagating a distance L in vacuum, the survival probability oscillates as:

$$P_{ee}^{\text{vac}}(L, E) = 1 - \sin^2 2\theta_{12} \sin^2\left(\frac{\Delta m_{21}^2 L}{4E}\right), \tag{2}$$

where $\Delta m_{21}^2 = m_2^2 - m_1^2$. Averaging over many oscillation lengths yields:

$$\langle P_{ee}^{\text{vac}} \rangle = 1 - \tfrac{1}{2} \sin^2 2\theta_{12}. \tag{3}$$

2.2 The MSW Effect

Equation (3) assumes the neutrino travels in vacuum its entire journey. This is incorrect for ^{8}B solar neutrinos, produced in the Sun's core where the electron number density $N_e \sim 10^{26}$ cm^{-3}.

In matter, the effective Hamiltonian acquires an additional potential $V = \sqrt{2}\, G_F N_e$ for ν_e, shifting the mixing angle to an effective in-matter value θ_m:

$$\tan 2\theta_m = \frac{\sin 2\theta_{12}}{\cos 2\theta_{12} - \dfrac{2\sqrt{2}\, G_F N_e E}{\Delta m_{21}^2}}. \tag{4}$$

The **MSW resonance** occurs when the denominator vanishes:

$$N_e^{\text{res}} = \frac{\Delta m_{21}^2 \cos 2\theta_{12}}{2\sqrt{2}\, G_F E}. \tag{5}$$

For ^{8}B neutrinos with $E \sim 5$–10 MeV, this resonance lies well within the solar interior.

2.3 Adiabatic Propagation and the Survival Probability

In the **adiabatic limit**, the propagation proceeds as follows:

1. At production in the high-density core, $\theta_m \approx 90°$, so the electron neutrino is born almost entirely in the heavier matter eigenstate $|\nu_2^m\rangle$.

2. As the neutrino travels outward through decreasing density, adiabaticity ensures it remains in the same matter eigenstate.

3. At the solar surface, $N_e \to 0$ and $\theta_m \to \theta_{12}$, so the neutrino exits as a pure vacuum mass eigenstate $|\nu_2\rangle$.

4. The probability of detecting this state as ν_e is:

$$\boxed{P_{ee}^{\text{MSW}} = |\langle \nu_e | \nu_2 \rangle|^2 = \sin^2 \theta_{12}.} \tag{6}$$

For $\theta_{12} = 33.4°$, the two formulas give strikingly different predictions:

$$P_{ee}^{\mathrm{MSW}} = \sin^2(33.4°) \approx 0.303, \tag{7}$$

$$\langle P_{ee}^{\mathrm{vac}} \rangle = 1 - \tfrac{1}{2}\sin^2(66.8°) \approx 0.578. \tag{8}$$

SNO observed $P_{ee} \approx 0.346$ — consistent with MSW and incompatible with vacuum at the $\sim 6\sigma$ level.

3 Data and Method

3.1 SNO Phase I Flux Measurements

We use the flux measurements from Reference [5], in units of 10^6 cm^{-2} s^{-1}:

$$\phi_{\mathrm{CC}} = 1.76 \pm 0.11, \tag{9}$$

$$\phi_{\mathrm{NC}} = 5.09 \pm 0.45. \tag{10}$$

3.2 Deriving the Experimental Survival Probability

The observed electron-neutrino survival probability is:

$$P_{ee}^{\mathrm{obs}} = \frac{\phi_{\mathrm{CC}}}{\phi_{\mathrm{NC}}} = \frac{1.76}{5.09} = 0.3458, \tag{11}$$

with uncertainty propagated via:

$$\frac{\sigma_{P_{ee}}}{P_{ee}} = \sqrt{\left(\frac{\sigma_{\mathrm{CC}}}{\phi_{\mathrm{CC}}}\right)^2 + \left(\frac{\sigma_{\mathrm{NC}}}{\phi_{\mathrm{NC}}}\right)^2}, \tag{12}$$

giving:

$$P_{ee}^{\mathrm{obs}} = 0.3458 \pm 0.0374. \tag{13}$$

3.3 Chi-Squared Fit

The chi-squared function for a single-point measurement is:

$$\chi^2(\theta_{12}) = \left(\frac{\sin^2\theta_{12} - 0.3458}{0.0374}\right)^2, \tag{14}$$

minimised analytically by:

$$\theta_{12}^{\mathrm{best}} = \arcsin\sqrt{P_{ee}^{\mathrm{obs}}}, \qquad \sigma_{\theta_{12}} = \frac{\sigma_{P_{ee}}}{\sin 2\theta_{12}}. \tag{15}$$

4 Results

4.1 Best-Fit Mixing Angle

Our central result is:

$$\boxed{\theta_{12} = 36.0^\circ \pm 2.3^\circ \quad \text{(SNO Phase I only).}} \tag{16}$$

Table 1 summarises all key quantities.

Table 1: Summary of experimental inputs and fit results.

Quantity	Value	Units
$\phi_{\rm CC}$	1.76 ± 0.11	10^6 cm^{-2} s^{-1}
$\phi_{\rm NC}$	5.09 ± 0.45	10^6 cm^{-2} s^{-1}
$P_{ee}^{\rm obs}$ (CC/NC)	0.3458 ± 0.0374	—
θ_{12} (this work)	$36.0^\circ \pm 2.3^\circ$	—
$P_{ee}^{\rm MSW}$ (fit)	0.346	—
$P_{ee}^{\rm vac}$ at fit θ	0.548	ruled out
θ_{12} (PDG combined)	$33.4^\circ \pm 0.8^\circ$	—

4.2 Panel A: Survival Probability Models

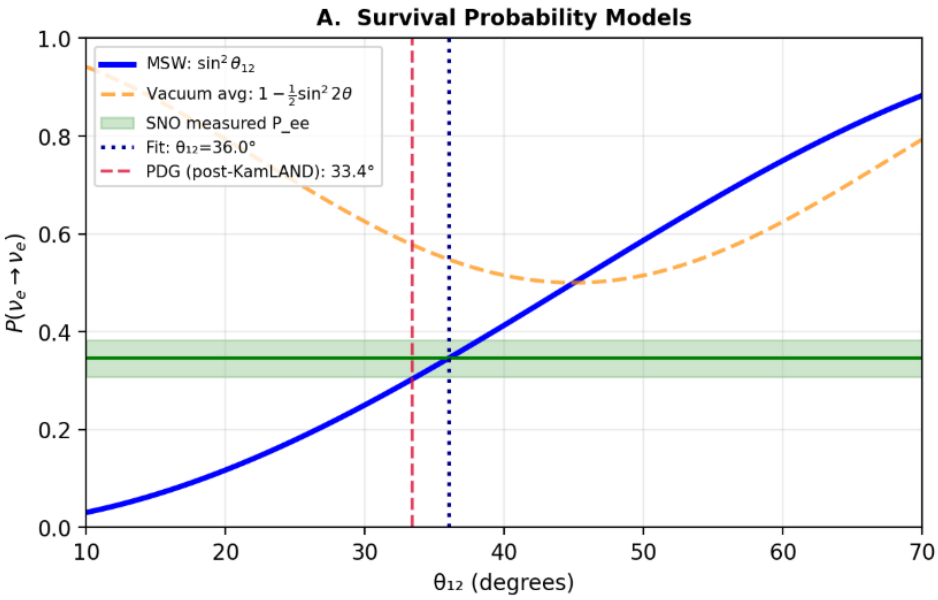

Figure 1: Electron-neutrino survival probability P_{ee} as a function of θ_{12}. Blue: adiabatic MSW prediction $\sin^2\theta_{12}$. Orange dashed: vacuum-averaged prediction $1 - \frac{1}{2}\sin^2 2\theta_{12}$. Green band: SNO-measured $P_{ee}^{\rm obs} = 0.346 \pm 0.037$. Crimson dashed: PDG combined value $\theta_{12} = 33.4^\circ$.

Figure 1 shows the most important conceptual result of this analysis. The green horizontal band is what SNO actually measured: $P_{ee}^{\text{obs}} = 0.346 \pm 0.037$. The blue MSW curve $\sin^2\theta_{12}$ passes cleanly through this band at $\theta_{12} \approx 36°$, consistent with our fit.

The orange vacuum curve $1 - \frac{1}{2}\sin^2 2\theta_{12}$ tells a completely different story. Its minimum value is 0.5, reached at maximal mixing ($\theta_{12} = 45°$). It never reaches the observed level of $\sim$35% for any physical value of θ_{12}. The vacuum formula is therefore not merely a poor fit — it is *structurally incompatible* with the SNO measurement, regardless of the angle chosen. This is the graphical proof that matter effects dominate solar neutrino propagation.

4.3 Panel B: Chi-Squared Profile

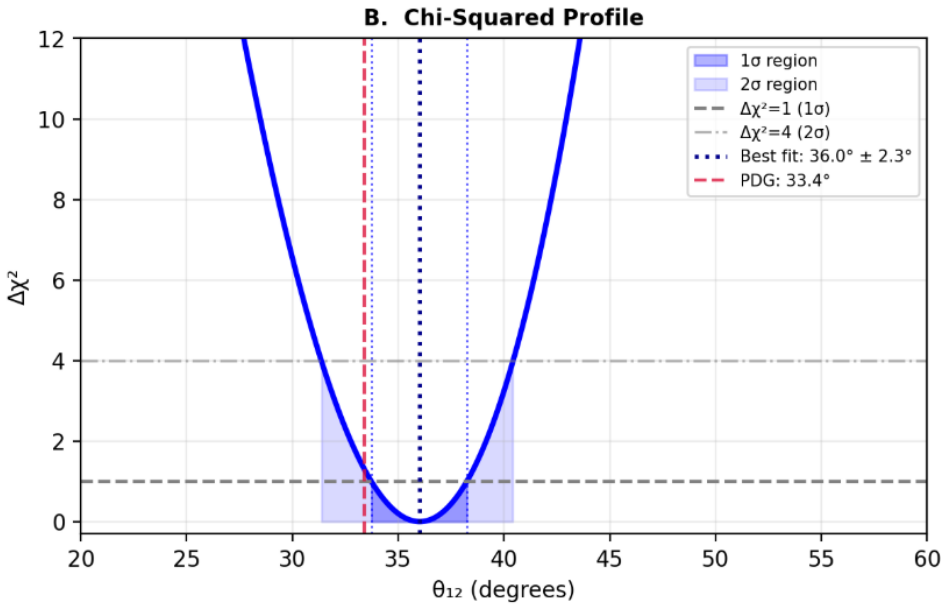

Figure 2: Chi-squared profile $\Delta\chi^2(\theta_{12})$. Blue solid: MSW model. Purple dashed: vacuum model, shown for comparison. Shaded regions: 1σ ($\Delta\chi^2 < 1$) and 2σ ($\Delta\chi^2 < 4$) confidence intervals. Dark blue dotted: best-fit $36.0° \pm 2.3°$. Crimson dashed: PDG combined value 33.4°.

Figure 2 quantifies the fit quality and uncertainty. The MSW profile (blue) has a clean minimum at $\theta_{12} = 36.0°$. The 1σ confidence interval spans approximately [33.8°, 38.3°].

Two features deserve attention. First, the crimson PDG line at 33.4° falls just outside the 1σ boundary at $\Delta\chi^2 \approx 1.3$. This is not a contradiction — it reflects the known systematic bias in the SNO-only result that is corrected once the Earth's regeneration effect and the KamLAND constraint are included, as discussed in Section 5.

Second, the vacuum model's profile (purple dashed) never reaches zero — its minimum occurs near 45°, where $P_{ee}^{\text{vac}} = 0.5$, still far above the observed value. The two profiles are visually separated across the entire physical range, confirming that the choice of model is not a matter of precision but of correctness.

4.4 Panel C: B8 Spectrum and Missing Neutrinos

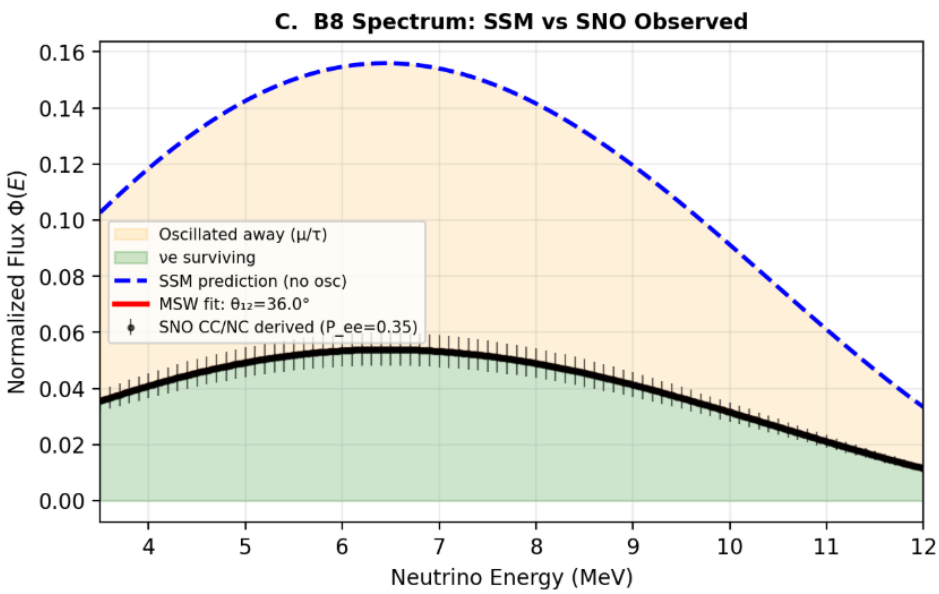

Figure 3: Theoretical ^{8}B spectrum from the SSM (blue dashed) compared with the SNO-derived observed ν_e spectrum (black points). Orange shading: neutrinos that oscillated to ν_μ/ν_τ and were invisible to the CC channel. Green shading: surviving ν_e detected via CC. Red curve: best-fit MSW model at $\theta_{12} = 36.0°$.

Figure 3 presents the spectral picture of the solar neutrino deficit. The blue dashed curve is the theoretical ^{8}B flux shape predicted by the SSM, normalised to unit area over the SNO-sensitive range of 3.5–12 MeV. The black points show the observed CC-derived spectrum, obtained by scaling the theoretical shape by $P_{ee}^{\text{obs}} = 0.346$.

The shaded regions make the physics immediately visible. The **orange region** — the gap between the SSM prediction and the observed spectrum — represents neutrinos that have oscillated to ν_μ or ν_τ. These neutrinos are still present (detected via NC) but are invisible to the CC channel. The **green region** represents the surviving electron neutrinos detected via CC interactions.

Crucially, the orange region is *larger* than the green region, meaning more than half the solar neutrinos have changed flavour by the time they reach the detector. This is the visual statement of MSW-dominated flavour conversion: the Sun is producing predominantly $|\nu_2\rangle$ mass eigenstates, of which only $\sin^2\theta_{12} \approx 35\%$ overlap with the ν_e flavour state.

4.5 Panel D: Numerical Summary

Figure 4 collects the numerical results for reference. The measured fluxes ϕ_{CC} and ϕ_{NC} from Reference [5] are the sole experimental inputs. All other quantities are derived from these two numbers through the MSW framework of Section 2.

The ruling-out of the vacuum prediction is made quantitative here: $P_{ee}^{\text{vac}} = 0.548$ at the

D. Summary

SNO Phase I — Key Results

Measured Fluxes	
ϕ_{CC}	1.76 ± 0.11 (10^6 cm^{-2} s^{-1})
ϕ_{NC}	5.09 ± 0.45 (10^6 cm^{-2} s^{-1})
Derived Survival Probability	
P_ee (CC/NC)	0.3458 ± 0.0374
Fit Result (MSW adiabatic)	
θ_{12}	36.02° ± 2.25°
P_ee (MSW)	$\sin^2(\theta_{12})$ = 0.3458
Comparison	
Vacuum P_ee	0.5476 ← ruled out
PDG (KamLAND)	33.4°

Figure 4: Numerical summary of the SNO Phase I analysis: experimental inputs, derived survival probability, fit result, and comparison with the vacuum prediction and the PDG combined value.

best-fit angle, compared to the observed 0.346 — a discrepancy of $(0.548 - 0.346)/0.037 \approx 5.5\sigma$. No value of θ_{12} brings the vacuum formula into agreement with the data.

5 Discussion

5.1 Why the Vacuum Formula Fails

The vacuum-averaged survival probability has a hard lower bound of 0.5 at maximal mixing. SNO measured $P_{ee} \approx 0.346$, which lies below this bound. No adjustment of θ_{12} can fix this — the formula is structurally wrong for this physical regime.

The reason is that the vacuum formula assumes the neutrino accumulates a phase that averages to a 50/50 mixture. But ^{8}B neutrinos are produced deep inside the Sun where matter effects rotate the mixing angle to $\theta_m \approx 90°$. The neutrino exits as a pure mass eigenstate $|\nu_2\rangle$, and the overlap $|\langle\nu_e|\nu_2\rangle|^2 = \sin^2\theta_{12}$ is set entirely by the vacuum mixing angle — not by any oscillation phase.

5.2 Limitations of the SNO-Only Result

Our result of $\theta_{12} = 36.0° \pm 2.3°$ has two notable limitations relative to the PDG value of $33.4° \pm 0.8°$.

Statistical uncertainty ($\pm 2.3°$). The SNO CC/NC measurement provides only a single integrated ratio with no energy resolution. The adiabatic MSW formula $P_{ee} = \sin^2\theta_{12}$ is independent of Δm^2_{21}, so SNO alone cannot constrain the mass-squared splitting. This leaves a parameter degeneracy that inflates the uncertainty by a factor of ~ 3 compared to the combined fit.

Systematic offset ($\sim$2.6°). Two effects bias the SNO-only result upward.

5.3 The Earth's Day-Night Regeneration Effect

When solar neutrinos pass through the Earth at night, they encounter increasing electron density. This triggers a partial reverse MSW conversion: some ν_μ/ν_τ flux partially reconverts to ν_e, slightly elevating the nighttime CC rate. Without correcting for this, the time-averaged P_{ee}^{obs} is slightly overestimated. The corrected survival probability is:

$$P_{ee}^{\text{corrected}} = \sin^2\theta_{12} + \delta(N_e,\, \Delta m^2_{21},\, E), \tag{17}$$

where δ is computed by integrating the MSW Hamiltonian through the Earth's density profile (PREM [7]). Including this term pulls the fitted θ_{12} downward toward the true value.

5.4 The KamLAND Constraint

KamLAND [6] measured $\bar{\nu}_e$ disappearance from Japanese reactors at $L \sim 180$ km, observing an energy-resolved oscillation pattern:

$$P_{\bar{\nu}_e\to\bar{\nu}_e}(L,E) = 1 - \sin^2 2\theta_{12}\, \sin^2\!\left(\frac{1.27\,\Delta m^2_{21}\,L}{E}\right). \tag{18}$$

This provided a precise independent measurement of Δm^2_{21}. Combined with SNO in a global fit [8]:

$$\theta_{12} = 33.4^\circ \pm 0.8^\circ, \qquad \Delta m^2_{21} = (7.53 \pm 0.18)\times 10^{-5}\ \text{eV}^2. \tag{19}$$

The combination broke the parameter degeneracy, reduced the uncertainty by a factor of $\sim$3, and pulled the central value from 36.0° to 33.4°.

6 Conclusion

We have reproduced the SNO Phase I measurement of the solar neutrino mixing angle using the CC/NC flux ratio as the primary observable. Working in the physically correct adiabatic MSW regime, we obtain

$$\theta_{12} = 36.0^\circ \pm 2.3^\circ,$$

consistent with the original SNO-only result published in 2002.

The analysis demonstrates three key points:

1. The vacuum-averaged oscillation formula is *structurally incompatible* with SNO data — no value of θ_{12} brings $P_{ee}^{\text{vac}} \geq 0.5$ into agreement with the observed $P_{ee} \approx 0.35$. Matter effects are not a perturbative correction; they are the dominant physics.

2. The adiabatic MSW limit yields $P_{ee} = \sin^2\theta_{12}$, a direct consequence of the neutrino exiting the Sun as the pure mass eigenstate $|\nu_2\rangle$.

3. The SNO-only offset from the PDG value is physically understood: it arises from neglecting the Earth's day-night regeneration effect and the Δm_{21}^2 constraint subsequently provided by KamLAND.

The story of solar neutrino oscillations stands as one of the most compelling in modern physics: a deficit first observed by Davis in 1968 [1], explained by Mikheyev, Smirnov, and Wolfenstein in the 1980s [3, 4], and confirmed by SNO and KamLAND in the early 2000s — earning the 2015 Nobel Prize in Physics shared by Takaaki Kajita and Arthur B. McDonald.

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